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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Retail Banking Transaction Analysis

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Project Overview

- A retail bank serves customers through accounts, cards and loans across 40 branches.
- The bank handles thousands of customer transactions and loan payments.
- Management wants data-driven answers about customers, account activity, transaction patterns, loan performance and product usage.
- As a Data Analyst supporting the Retail Banking team, we built the database in MySQL, analysed the data with SQL and recorded business insights.

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Business Scenario

- The Customer team needs to understand who the customers are and how segments differ.
- Management wants to see how accounts are used and how branches perform.
- Operations wants to understand how money moves through transactions and channels.
- The Loans team wants to know whether customers repay on time.
- The Cards team wants to understand card usage and product engagement.

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Business Objectives

- Understand the customer profile and segmentation.
- Understand account usage and branch activity.
- Analyze transaction patterns across types, channels and time.
- Evaluate loan performance and repayment behaviour.
- Understand card usage and product engagement across the customer base.

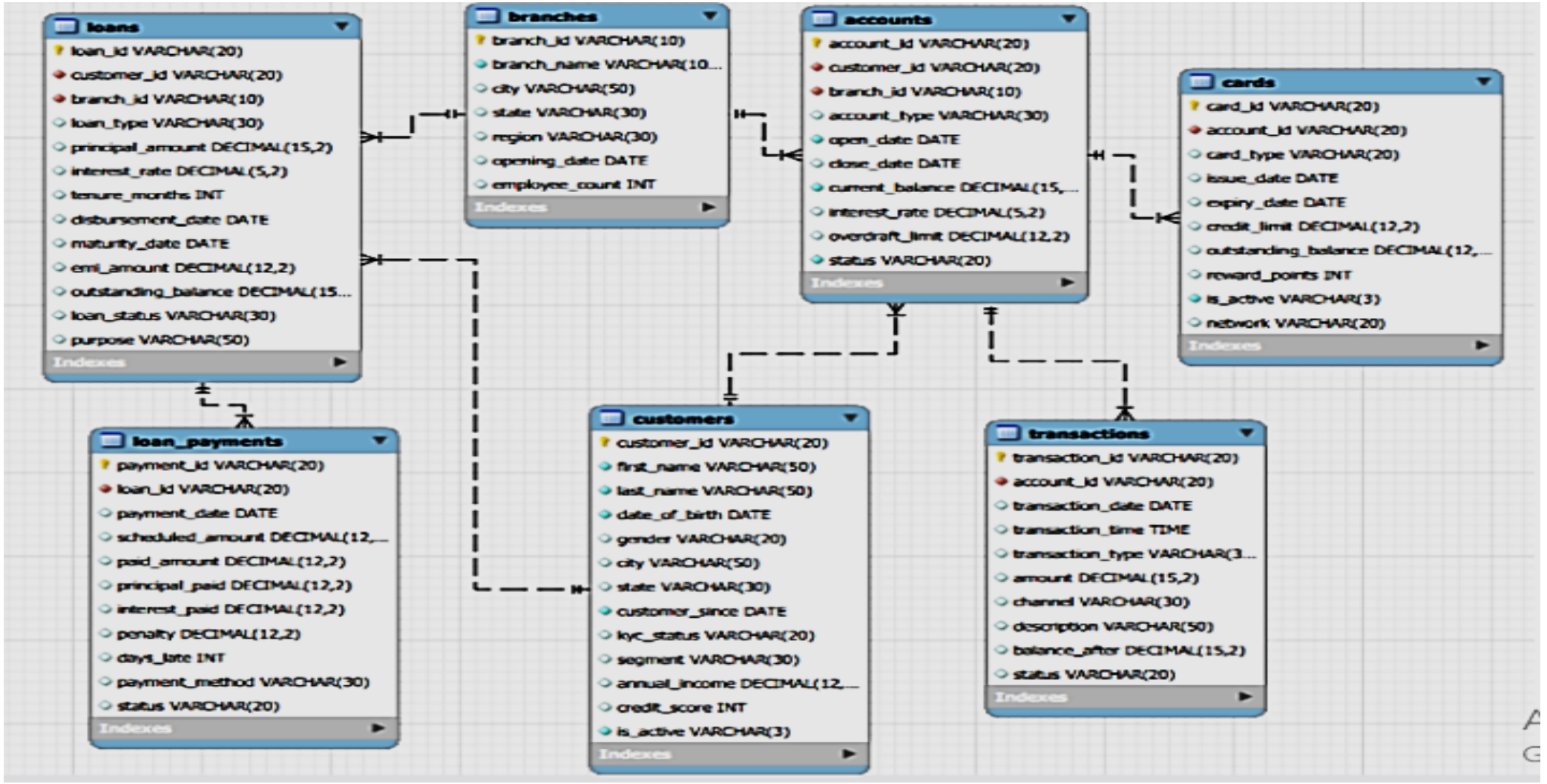
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Dataset & Database Overview

Table Name	No. of Rows	No. of Columns
customers	500	13
accounts	700	10
branches	40	7
loans	300	13
loan_payments	2,000	11
cards	600	10
transactions	5,000	10

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ER Diagram & Database Structure



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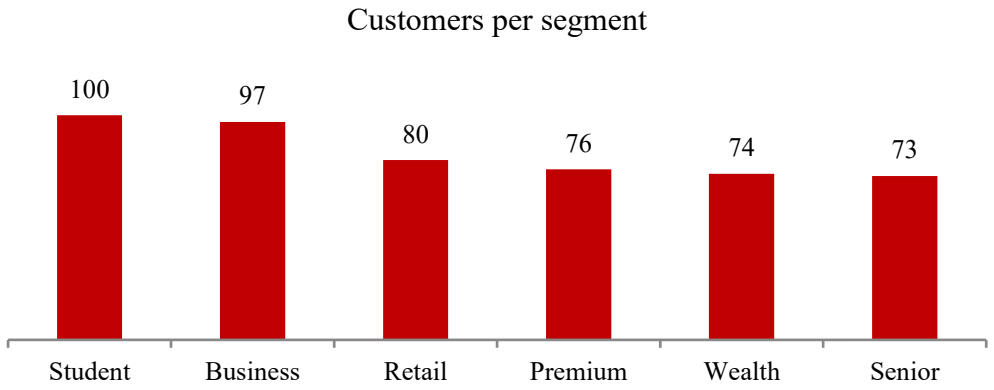
Basic Analysis

Question	Result
Total customers	500
Total accounts	700 (1.4 per customer)
Account types	6: Business Checking, CD, Checking, IRA, Money Market, Savings
Active customers	375 of 500 (75.0%)
Transaction types	6: ATM Withdrawal, Credit, Debit, Fee, Interest, Transfer
Completed transactions amount	$10,976,552.36 (1,688 transactions)
Loan types	6: Auto, Business, Credit Builder, Education, Home, Personal
Total loans	300
Card types	2: Credit, Debit
Outstanding loan balance	$53,441,891.25 (74.0% of $72.21M lent)

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4.1 Customer Profile & Segmentation

Output:



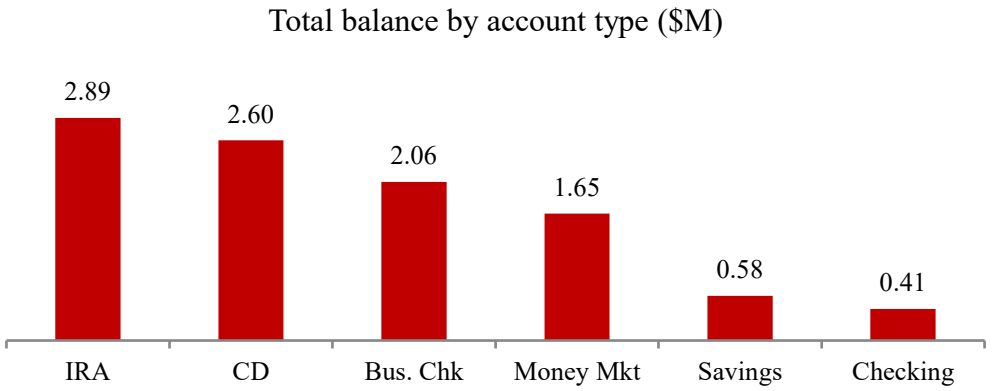
Key Insights:

- **Segments:** Student is largest (100, 20%); average income is $140K-$159K and credit score 679-687 in every segment, so labels do not reflect customer value. Redefine segments.
- **Geography & demographics:** No city has more than 3% of customers (the top 10 hold 27%). Gender is balanced. 30-44 is the largest age group (33.6%) and Under 30 is only 14%.
- **KYC risk:** Only 28.6% of customers are fully verified; 241 of 375 active customers (64.3%) have Partial/Pending KYC. Run a KYC completion drive.
- **Tenure:** Average tenure is about 7.4 years; Premium (6.8) and Wealth (6.9) are the newest customers.

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4.2 Account Usage & Branch Activity

Output:



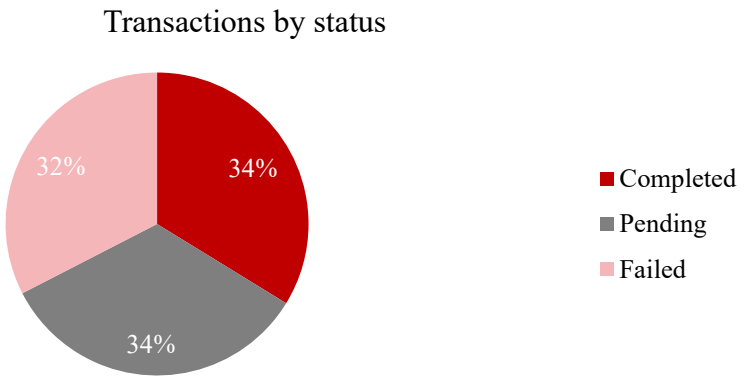
Key Insights:

- **Deposits:** $10.2M in total; IRA, CD and Money Market hold 70% at higher rates (3.3%-4.6%). Savings and Checking are popular but small.
- **Relationships:** Customers hold 1.4 accounts on average. One customer has 8 accounts but a zero balance, while others hold several accounts with 100K+ balances.
- **Branches:** BR025 holds $501,926 and BR010 only $58,688. More accounts does not mean more money: BR011 has the most accounts (28) but only $306K.
- **Closures:** 203 of 700 accounts (29%) are closed and hold zero balance. No account is overdrawn. Add retention offers.

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4.3 Transaction Patterns

Output:



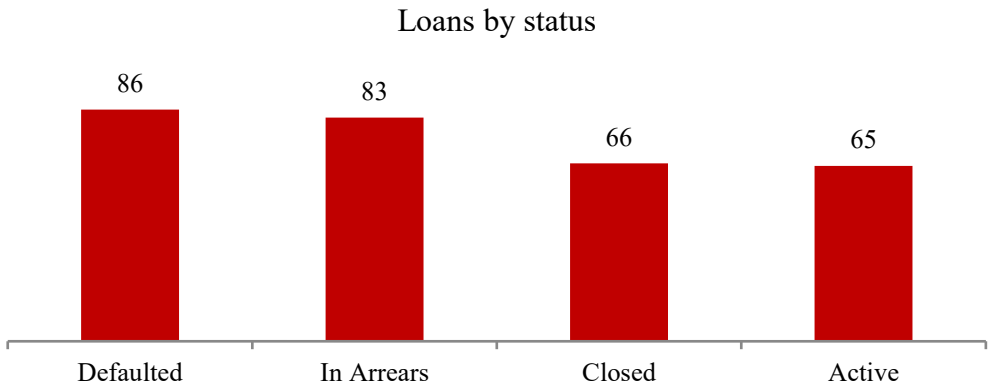
Key Insights:

- **Completion problem:** Only 33.8% of transactions complete; 33.7% are Pending and 32.6% Failed.
- **Channels & types:** Online and Mobile carry 49.9% of activity, Branch 22.9%; types are evenly spread. Fee and Interest averages (~$7,000) need validation.
- **Usage:** Investment, Utility and Rent dominate; Salary is only 5.6% (payroll opportunity). Business is the most active segment, Premium the least.
- **Trend:** Volume grew 1,279 (2022) to 1,723 (2023) to 1,933 (2024), but growth is slowing; there is no data for Jan 2025.

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4.4 Loan Performance & Repayment

Output:



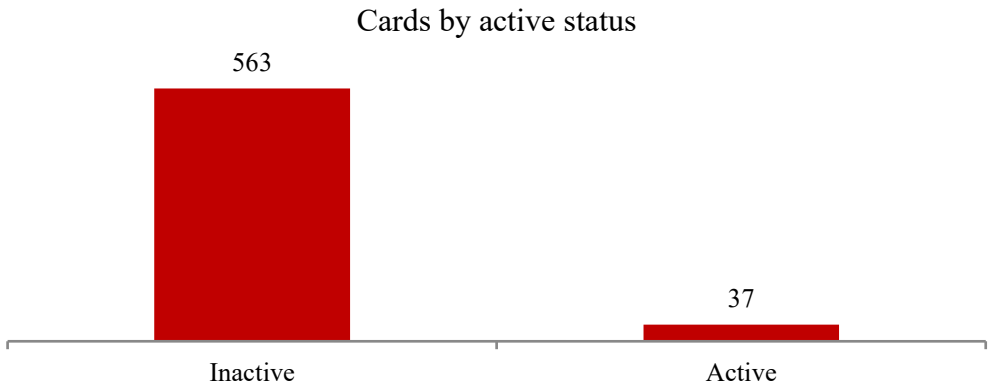
Key Insights:

- **Loan book:** $53.44M of $72.21M (74%) is still outstanding; 169 loans (56.3%) are Defaulted or In Arrears, holding 73.3% of balances. Education, Business, Auto and Home are about 70% of lending.
- **Repayment:** One in four payments (25%) is late across all loan types (21.7% to 28.2%). Penalties total $93,961, under 0.2% of the money owed.
- **Payment method:** Cheque, Bank Transfer and Cash are about 57% of payments. Auto Debit is only 5.6%. Promote Auto Debit.
- **Branches:** The late-payment share runs from 5.9% (BR037) to 45.5% (BR012). Review the worst branches.

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4.5 Card Usage & Product Engagement

Output:



Key Insights:

- **Cards:** 93.8% of cards are inactive, credit utilization is healthy at 41.8%.
- **Rewards:** Debit and Credit earn almost the same points; Visa holds the most points (28%).
- **Engagement:** Only 124 customers (24.8%) use accounts, cards and loans together. 267 customers hold a card, and 124 of them also have a loan.
- **Credit cards:** 242 cards (40%), average limit about $24.3K, and customers owe $2.46M.

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Overall Key Findings

1. **Customers**
   500 customers, 75% active, spread over many cities. Only 28.6% fully KYC-verified.

2. **Accounts**
   $10.2M in deposits, concentrated in IRA, CD and Money Market; 29% of accounts closed.

3. **Transactions**
   Growing yearly and led by Online/Mobile, but only 33.8% complete.

4. **Loans**
   56.3% of loans are Defaulted or In Arrears, holding 73.3% of the $53.4M still owed. One in four payments is late.

5. **Cards**
   93.8% of cards inactive. Only 124 customers use account, card and loan together.

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Business Recommendations

- **Collections:** Follow up the 169 defaulted and in-arrears loans first (73% of the money owed). Tighten approval for Home and Business loans.
- **Transactions:** Fix processing failures; two thirds of transactions fail or stay pending.
- **KYC:** Complete KYC for the 241 active Partial/Pending customers.
- **Repayment:** Push Auto Debit (only 5.6% of payments) and send pre-due reminders.
- **Cards & cross-sell:** Reactivate or close the 563 inactive cards. Offer cards to the 233 customers with none and loans to the 143 card holders with no loan.
- **Operations:** Redefine segments using income and credit bands. Review weak branches (BR010, BR017, BR022 for balances and BR012, BR020 for late payments).

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Data Quality Notes

- close_date was blank for 497 open accounts; loaded through a staging table using NULLIF().
- Primary and foreign key constraints were enforced during import, so there are no duplicate keys or orphan records.
- Only a third of transactions are completed, so totals include pending and failed ones. Fee and Interest averages look large (about $7K).
- 98.7% of transactions fall in 2022-2024 and none in Jan 2025; customer segments differ very little in income or credit score.

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Conclusion

- SQL analysis turned 7 tables into clear answers on customers, accounts, transactions, loans and cards.
- The customer base is balanced and active, but segment labels are weak and KYC completion is the most urgent compliance issue.
- The biggest risks are the troubled loan book (56.3% Defaulted or In Arrears) and a transaction completion rate of only 33.8%.
- Card engagement is weak but cross-sell potential is large, supporting data-driven decisions to improve retail banking performance.

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THANK YOU

